

QIMING CUI

Applied Mathematics and Statistics Dept., Johns Hopkins University, Maryland, USA 21218

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SUMMARY

My PhD research specializes in algorithms with machine-learned predictions (learning-augmented algorithms), algorithmic decision-making under uncertainty, online algorithms, and optimization. Generally, I am interested in applying quantitative analysis and mathematical methods to solve complex problems.

EDUCATIONAL BACKGROUND

Johns Hopkins University, USA

Doctor of Philosophy in Applied Math and Statistics

May 2027 (expected)

Adviser: Dr. Michael Dinitz

University of Oxford, UK

Master of Science in Mathematical Sciences

July 2022

Graduate with Merit

University of Liverpool, UK

Bachelor of Science with Honours in Mathematics (First Class)

June 2021

GPA: 4.0/4.0

RELEVANT COURSEWORK

- ML: Machine Learning Theory, Probability Theory, Statistics Theory, Reinforcement Learning, Large Language Models
- Optimization: Convex Optimization, Continuous Optimization, Nonlinear Optimization, Combinatorial Optimization
- Algorithms: Algorithm Design and Analysis, Approximation Algorithms, Online Algorithms, Differential Privacy

PROFESSIONAL SKILLS

- Programming and Data Tools: Python (NumPy, Pandas, SciPy, Scikit-learn), SQL, MATLAB, Excel, LaTeX
- Machine Learning and AI Tools: ChatGPT, Claude, LLM agents and tool use, PyTorch, Hugging Face

RESEARCH PROJECT EXPERIENCES

Contention resolution with distributional predictions

- Submitted the paper to SIAM Symposium on Simplicity in Algorithms (SOSA 2027)
- Developed new algorithms and theoretical guarantees that generalize and improve upon prior state-of-the-art results
- Proved the first prediction error dependent lower bound and showed it is essentially matching the above guarantees

Ski rental with distributional predictions of unknown quality

- Published and presented the paper at the International Conference on Machine Learning (ICML 2026)
- Designed the first robust algorithm that leverages full distributional predictions with unknown prediction error
- Proved that the algorithm has better performance than the state-of-the-art algorithm given distributional predictions
- Proved that the algorithm is essentially the best by giving consistency-robustness guarantees and tight lower bounds

Controlling tail risk in two-slope ski rental

- Published and presented the paper at Approximation and Online Algorithms (WAOA 2025) and invited to special issue
- Authored the first paper to develop technical theorems and characterize the structure of optimal solutions
- Designed two algorithms to compute the optimal solutions focusing on the speed and the accuracy respectively

β -expansions

- Finished a final year (2021) report about theorems and properties of β -expansions from dynamical systems viewpoint
- Exploited the number of G -expansions, established a new theorem and gave proofs of it
- Studied Uniqueness expansions of points, provided simpler proofs of existing theorems

ACADEMIC SERVICE

- Giving presentations in seminars/conferences: JHU Theory seminar, WAOA 25', ICML 26'
- Co-instructed the course EN.553.803 Financial Computing Workshop (Intro to Agentic AI) at JHU
- Working as a Teaching Assistant in Machine Learning (EN.553.740), Computing for Applied Math (EN.553.688), Approximation Algorithms (EN.601.635) and Numerical Linear Algebra (EN.553.385)

HONORS AND AWARDS

- 2021 Bronze medal in 28th International Mathematics Competition for University Students
- 2019-2020, 2020-2021 University of Liverpool Excellence Scholarship (1/320)
- 2019 Summer Undergraduate Research Fellowship

PERSONAL INTERESTS

- World travel, exploring new experiences, natural landscapes, cruises, punting
- Crime documentaries and case analysis, critical analysis of current affairs